

# Indices, Surds & Quadratics

## ANSWER KEY



### Laws of indices

- 1 Simplify  $x^5 \times x^{-2}$ .  
 $x^5 \times x^{-2} = x^{5-2} = x^3$ .
  - 2 Simplify  $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ .  
 $\left(\frac{8x^6}{27y^3}\right)^{1/3} = \frac{8^{1/3}x^{6/3}}{27^{1/3}y^{3/3}} = \frac{2x^2}{3y}$ .
  - 3 Simplify  $\frac{x^{-3} \times x^7}{x^2}$ , giving your answer as a single power of  $x$ .  
 $\frac{x^{-3} \times x^7}{x^2} = \frac{x^4}{x^2} = x^2$ .
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### Solving exponential equations

- 4 Solve  $3^{2x-1} = 81$ .  
Since  $81 = 3^4$ :  $2x - 1 = 4$ , so  $x = \frac{5}{2}$ .
  - 5 Solve  $4^x = 8^{x-1}$ , writing both sides with base 2 first.  
 $2^{2x} = 2^{3(x-1)}$ , so  $2x = 3x - 3$ , giving  $x = 3$ .
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### Simplifying surds

- 6 Simplify  $\sqrt{75}$ .  
 $\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$ .
  - 7 Simplify  $\sqrt{20} + \sqrt{45}$ .  
 $\sqrt{20} = 2\sqrt{5}$  and  $\sqrt{45} = 3\sqrt{5}$ , so the sum is  $2\sqrt{5} + 3\sqrt{5} = 5\sqrt{5}$ .
  - 8 Simplify  $\sqrt{48} - \sqrt{12} + \sqrt{3}$ .  
 $\sqrt{48} = 4\sqrt{3}$ ,  $\sqrt{12} = 2\sqrt{3}$ . Sum:  $4\sqrt{3} - 2\sqrt{3} + \sqrt{3} = 3\sqrt{3}$ .
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### Rationalizing the denominator

- 9 Rationalize  $\frac{6}{\sqrt{3}}$ .  
 $\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$ .
- 10 Rationalize  $\frac{5}{2-\sqrt{3}}$  by multiplying by the conjugate.  
 $\frac{5}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{5(2+\sqrt{3})}{4-3} = 5(2+\sqrt{3}) = 10 + 5\sqrt{3}$ .

11 Rationalize  $\frac{4}{\sqrt{7}+1}$

$$\frac{4}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1} = \frac{4(\sqrt{7}-1)}{7-1} = \frac{4(\sqrt{7}-1)}{6} = \frac{2(\sqrt{7}-1)}{3}.$$


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## Completing the square

- 12 Write  $x^2 - 8x + 10$  in the form  $(x - a)^2 - b$ , stating  $a$  and  $b$ .  
 $x^2 - 8x + 10 = (x^2 - 8x + 16) - 6 = (x - 4)^2 - 6$ . So  $a = 4$ ,  $b = 6$ .
- 13 Use completing the square to find the minimum value of  $f(x) = 2x^2 - 12x + 7$ .  
 $2x^2 - 12x + 7 = 2(x^2 - 6x) + 7 = 2(x^2 - 6x + 9) - 18 + 7 = 2(x - 3)^2 - 11$ . Minimum value is  $-11$ , occurring at  $x = 3$ .
- 14 Use completing the square to solve  $x^2 + 6x - 7 = 0$ .  
 $(x^2 + 6x + 9) - 9 - 7 = 0$ , so  $(x + 3)^2 = 16$ , giving  $x + 3 = \pm 4$ , so  $x = 1$  or  $x = -7$ .
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## Solving quadratics: factoring and the formula

- 15 Solve  $x^2 - 2x - 15 = 0$  by factoring.  
 Factor:  $(x - 5)(x + 3) = 0$ , so  $x = 5$  or  $x = -3$ .
- 16 Solve  $3x^2 + 2x - 4 = 0$  using the quadratic formula, giving exact answers.  
 $x = \frac{-2 \pm \sqrt{4 + 48}}{6} = \frac{-2 \pm \sqrt{52}}{6} = \frac{-2 \pm 2\sqrt{13}}{6} = \frac{-1 \pm \sqrt{13}}{3}.$
- 17 Solve  $x^2 - 6x + 9 = 0$ , and state what is special about the solution.  
 $(x - 3)^2 = 0$ , so  $x = 3$  is a repeated (double) root — the discriminant of this equation is zero.
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## The discriminant

- 18 Use the discriminant to determine the number of real roots of  $x^2 + 4x + 7 = 0$ .  
 Discriminant  $= 4^2 - 4(1)(7) = 16 - 28 = -12$ . Since this is negative, there are no real roots.
- 19 Find the values of  $k$  for which  $x^2 + kx + 9 = 0$  has exactly one (repeated) real root.  
 For a repeated root, the discriminant must equal zero:  $k^2 - 4(1)(9) = 0$ , so  $k^2 = 36$ , giving  $k = 6$  or  $k = -6$ .
- 20 Show that  $2x^2 + 3x + 5 = 0$  has no real roots, using the discriminant.  
 Discriminant  $= 3^2 - 4(2)(5) = 9 - 40 = -31$ . Since this is negative, the equation has no real roots.
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## Quadratic equations disguised in another variable

- 21** Solve  $x^4 - 13x^2 + 36 = 0$  by substituting  $u = x^2$ .  
 $u^2 - 13u + 36 = 0$ , factoring to  $(u - 4)(u - 9) = 0$ , so  $u = 4$  or  $u = 9$ . Since  $u = x^2$ :  $x = \pm 2$  or  $x = \pm 3$ .
- 22** Solve  $2^{2x} - 5(2^x) + 4 = 0$  by substituting  $u = 2^x$ .  
 $u^2 - 5u + 4 = 0$ , factoring to  $(u - 1)(u - 4) = 0$ , so  $u = 1$  or  $u = 4$ . Since  $u = 2^x$ :  $2^x = 1$  gives  $x = 0$ ;  $2^x = 4$  gives  $x = 2$ .

### Sum and product of the roots

- 23** For the quadratic  $2x^2 - 7x + 3 = 0$ , find the sum and product of its roots without solving for them, using  $\alpha + \beta = -\frac{b}{a}$  and  $\alpha\beta = \frac{c}{a}$ .  
Sum:  $\alpha + \beta = -\frac{-7}{2} = \frac{7}{2}$ . Product:  $\alpha\beta = \frac{3}{2}$ .
- 24** A quadratic equation has roots  $\alpha = 4$  and  $\beta = -3$ . Find a quadratic equation with these roots, in the form  $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ .  
 $\alpha + \beta = 1$ ,  $\alpha\beta = -12$ . Equation:  $x^2 - x - 12 = 0$ .

### A well-designed word problem: optimizing area

- 25** This question has two parts. A farmer has 60 meters of fencing to enclose a rectangular field against an existing straight wall (so fencing is only needed on three sides). Let the width perpendicular to the wall be  $x$  meters. (a) Show that the enclosed area is  $A = x(60 - 2x)$ . (b) Use completing the square to find the value of  $x$  that maximizes the area, and state the maximum area.
- (a) If the two widths perpendicular to the wall each use  $x$  meters of fencing, the remaining fencing for the side parallel to the wall is  $60 - 2x$  meters, so the area is  $A = x(60 - 2x)$ . (b) Expand:  
 $A = 60x - 2x^2 = -2(x^2 - 30x) = -2[(x - 15)^2 - 225] = -2(x - 15)^2 + 450$ . The maximum area is 450 square meters, occurring at  $x = 15$  meters.